Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2026

Centre Number

Candidate Number

Digital Technology

Assessment Unit AS 1
assessing
Approaches to System Development

SDT11

[SDT11]

FRIDAY 15 MAY, AFTERNOON

TIME

1 hour 30 minutes.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
You must answer the questions in the spaces provided.
Do not write outside the boxed area on each page or on blank pages.
Complete in blue or black ink only. **Do not write with a gel pen.**
Answer **all five** questions.

INFORMATION FOR CANDIDATES

The total mark for this paper is 100.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in Questions **2(d)(ii)** and **4(b)(ii)**.

15024

1 **(a)** The systems analyst conducts fact-finding during system development.

(i) List four other responsibilities of the systems analyst.

1. _____

_____

2. _____

_____

3. _____

_____

4. _____

_____ [4]

Interviews and questionnaires are two methods of fact-finding.

(ii) Describe two advantages of using interviews instead of questionnaires for fact-finding.

1. _____

_____

_____

_____

2. _____

_____

_____

_____ [4]

(iii) Describe two disadvantages of using interviews instead of questionnaires for fact-finding.

1. _____

_____

_____

_____

2. _____

_____

_____

_____ [4]

[Turn over



(b) Data flow diagrams are used during the analysis stage.

(i) Describe two main differences between a context level data flow diagram and a level 1 data flow diagram.

1. _____

_____

_____

_____

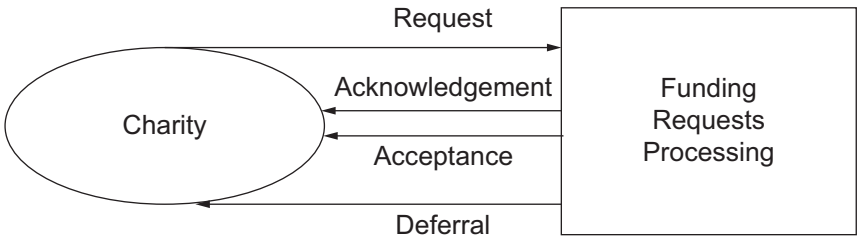
2. _____

_____

_____

_____ [4]

(ii) The following context level data flow diagram shows how requests from charities for funding are processed.



Details of all requests are held in a Requests file.

Each request is processed in two stages as follows:

Stage 1 The request is logged and an acknowledgement sent to the charity. The Requests file is updated.

Stage 2 The request is reviewed and either accepted or deferred. Notice of the decision is sent to the charity. The Requests file is updated.

In the box below, produce a level 1 data flow diagram for this system as described above.

[8]

[Turn over

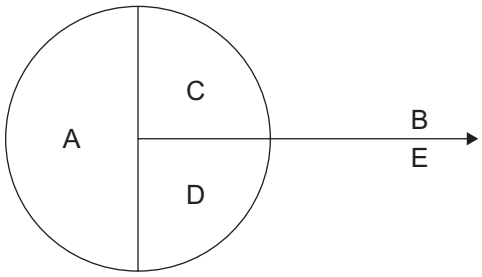
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2 **(a)** Critical path analysis is an important project management tool.

(i) A network diagram is produced as part of critical path analysis.

The notation for a node in a network diagram is shown below.



The following table lists the components of a node.

Complete the table by filling in the names of the three missing components.

Reference	Component
A	Node number
B	Task identifier
C	
D	
E	

[3]

(ii) Explain how the critical path assists in the management of a software project.

[5]

(b) With the aid of an example in each case, explain the difference between a constraint and a risk with regard to a software project.

Constraint

Example

Risk

Example

[6]

(c) Explain why version management of source code is important during system development.

[4]

[Turn over

(d) An information system can be developed using Rapid Application Development (RAD) or by using the waterfall method.

(i) Process modelling is one of the stages in RAD.

Describe the purpose of the process modelling stage.

_____

_____

_____

_____ [2]

(ii) Compare RAD and the waterfall method with respect to how each involves the end user.

Quality of written communication will be assessed in this question.

[6]

[Turn over

3 **(a)** A test plan was produced during the design of a stock control system.

(i) List four elements of a test plan.

1. _____

2. _____

3. _____

4. _____ [4]

(ii) One module of the stock control system performs a validation check on a stock level. This should be an integer in the range 1 to 1000 inclusive.

Explain what is meant by each of the following, giving an example of the test data which could be used with this module.

Extreme data

_____

_____

Example

_____

Exceptional data

_____

_____

Example

_____ [4]

(b) System testing and acceptance testing were used with the stock control system.

Compare the purpose of each of these types of testing.

System testing

Acceptance testing

[4]

[Turn over

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4 **(a)** An existing information system is being replaced by a new information system.

Explain how data conversion may be carried out in this case.

[4]

[Turn over

(b) An organisation has a disaster recovery plan.

(i) Describe three considerations a disaster recovery plan should address.

1. _____

_____

_____

_____

2. _____

_____

_____

_____

3. _____

_____

_____

_____ [6]

(ii) The organisation's backup procedure consists of an incremental backup every day followed by a full backup at the end of the week.

Describe each backup method and explain how the organisation's data will be recovered.

Quality of written communication will be assessed in this question.

[6]

[Turn over



5 **(a)** Programmers can use an Integrated Development Environment (IDE) to translate programs.

(i) Identify four other facilities provided by an IDE.

1. _____

_____

2. _____

_____

3. _____

_____

4. _____

_____ [4]

(ii) Programs can be translated using an interpreter or a compiler.

Describe two main differences between an interpreter and a compiler.

1. _____

_____

_____

_____

2. _____

_____

_____

_____ [4]

(b) The algorithm for calculating student grades in a test is shown below.

```
IF mark < 40
THEN
   grade = 'Fail'
END IF
IF mark >= 40 AND mark < 70
THEN
   grade = 'Pass'
END IF
IF mark >= 70
THEN
   grade = 'Distinction'
END IF
```

Rewrite the algorithm using a single **IF THEN ELSE** statement.

[10]

[Turn over



(c) Explain the following object-oriented programming terms.

Object

Method

[4]

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Question Number	**Marks**
1	
2	
3	
4	
5	
Total Marks	

Examiner Number

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